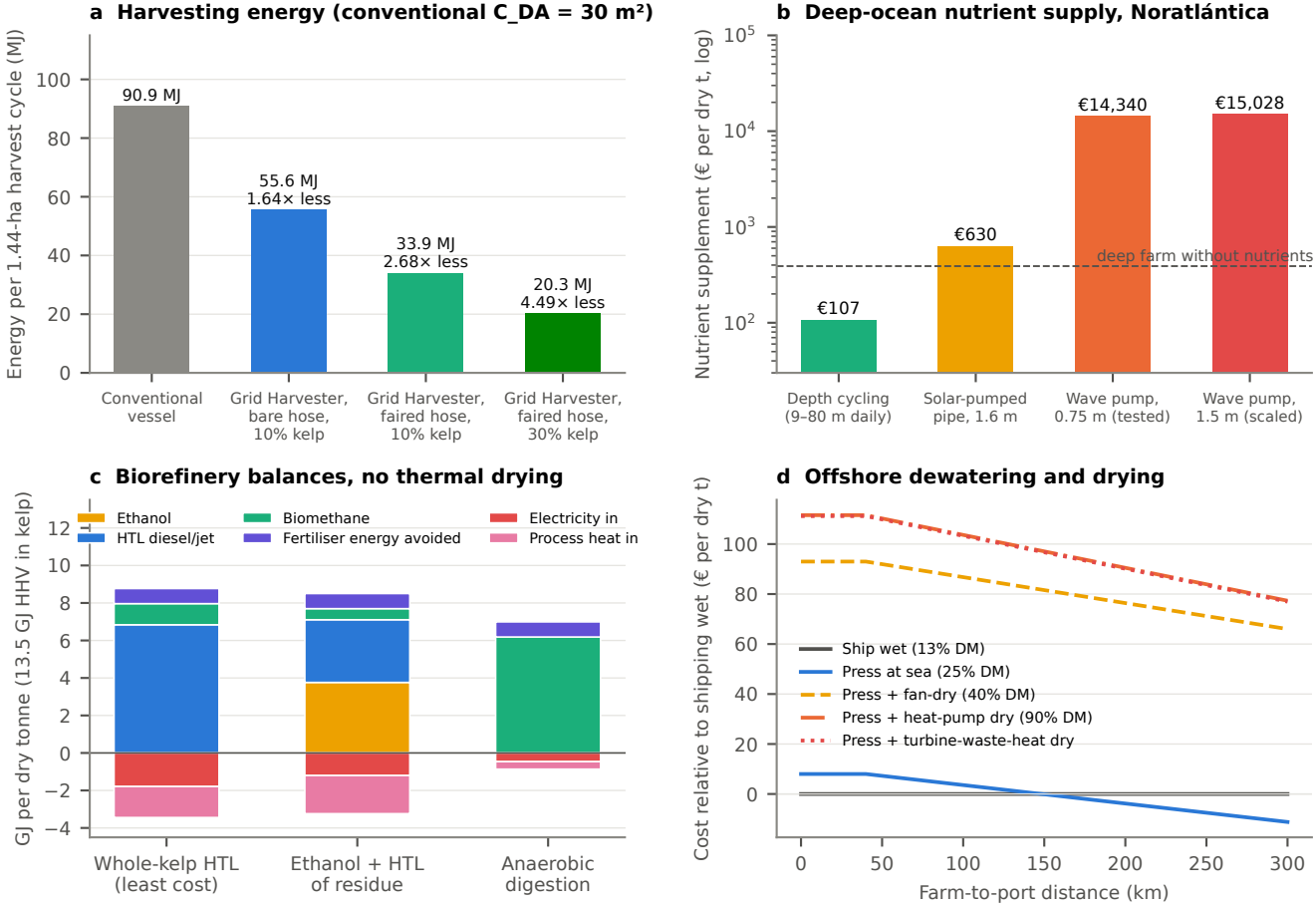
a Harvesting energy (conventional C_DA = 30 m²)
Energy per 1.44-ha harvest cycle (MJ)
90.9 MJ
55.6 MJ
1.64× less
33.9 MJ
2.68× less
20.3 MJ
4.49× less
Conventional vessel
Grid Harvester, bare hose, 10% kelp
Grid Harvester, faired hose, 10% kelp
Grid Harvester, faired hose, 30% kelp
b Deep-ocean nutrient supply, Noratlántica
Nutrient supplement (€ per dry t, log)
€107
€630
€14,340
€15,028
deep farm without nutrients
Depth cycling (9–80 m daily)
Solar-pumped pipe, 1.6 m
Wave pump, 0.75 m (tested)
Wave pump, 1.5 m (scaled)
c Biorefinery balances, no thermal drying
GJ per dry tonne (13.5 GJ HHV in kelp)
Ethanol
HTL diesel/jet
Biomethane
Fertiliser energy avoided
Electricity in
Process heat in
Whole-kelp HTL (least cost)
Ethanol + HTL of residue
Anaerobic digestion
d Offshore dewatering and drying
Cost relative to shipping wet (€ per dry t)
Ship wet (13% DM)
Press at sea (25% DM)
Press + fan-dry (40% DM)
Press + heat-pump dry (90% DM)
Press + turbine-waste-heat dry
Farm-to-port distance (km)